

Natural language statement

Domain

GroundTruth

2.31 Theorem. If λ is a left Haar measure on G and Δ is the modular function of G , then for every f in $L^1(G)$ $\int_G f(x^{-1})\Delta(x)^{-1}d\lambda(x) = \int_G f(x)d\lambda(x)$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_yf(x) = f(y^{-1}x)$ and $R_yf(x) = f(xy)$ are the left and right translates of f , and $f * g(x) = \int_G f(y)g(y^{-1}x)dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. The modular function $\Delta : G \rightarrow (0, \infty)$ is determined by $\lambda(\Delta x) = \Delta(x)\lambda(x)$ for a left Haar measure λ ; it is a continuous homomorphism and is identically 1 exactly when G is unimodular.

Harmonic analysis

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theorem folland_2_31_modular_inversion_formula {G : Type*} [Group G]
[TopologicalSpace G] [IsTopologicalGroup G]
[LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G]
(mu : Measure G) [mu.IsHaarMeasure]
(f : G -> C) (hf : Integrable f mu) :
integral x, f x^-1 * ((Measure.modularCharacterFun x : R>=0) : R) dmu =
integral x, f x dmu := by
sorry
```

2.45 Theorem. Let I be a closed subspace of $L^1(G)$. Then I is a left ideal if and only if it is closed under left translations, and I is a right ideal if and only if it is closed under right translations. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_yf(x) = f(y^{-1}x)$ and $R_yf(x) = f(xy)$ are the left and right translates of f , and $f * g(x) = \int_G f(y)g(y^{-1}x)dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution.

Harmonic analysis

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def leftTranslate (y : G) (f : G -> C) : G -> C := fun x => f (y^-1 * x)

def rightTranslate (y : G) (f : G -> C) : G -> C := fun x => f (x * y)

noncomputable def groupConv (mu : Measure G) (f g : G -> C) : G -> C :=
fun x => integral y, f y * g (y^-1 * x) dmu

def IsLpClosed (p : R>=0inf) (mu : Measure G) (I : Set (G -> C)) : Prop :=
forall f : G -> C, MemLp f p mu ->
(forall eps : R>=0inf, 0 < eps -> exists g in I, MemLp g p mu /\
eLpNorm (f - g) p mu < eps) -> f in I

theorem folland_2_45_closed_ideals_are_translation_invariant {G : Type*}
[Group G]
[TopologicalSpace G] [IsTopologicalGroup G]
[LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G]
(mu : Measure G) [mu.IsHaarMeasure]
(I : Submodule C (G -> C)) (hclosed : IsLpClosed 1 mu (I : Set (G -> C)))
(hmem : forall f in I, Integrable f mu) :
((forall g : G -> C, Integrable g mu -> forall f in I, groupConv mu g f in I) <=>
forall (y : G), forall f in I, leftTranslate y f in I) /\
((forall g : G -> C, Integrable g mu -> forall f in I, groupConv mu f g in I) <=>
forall (y : G), forall f in I, rightTranslate y f in I) := by
sorry
```

2.51 Theorem. Suppose G is a locally compact group and H is a closed subgroup. There is a G -invariant Radon measure μ on G/H if and only if $\Delta_G|_H = \Delta_H$. In this case, μ is unique up to a constant factor, and if this factor is suitably chosen we have $\int_G f(x)dx = \int_{G/H} \int_H f(xh)dxhd\mu(xH)$ for f in $C_c(G)$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_yf(x) = f(y^{-1}x)$ and $R_yf(x) = f(xy)$ are the left and right translates of f , and $f * g(x) = \int_G f(y)g(y^{-1}x)dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. $Pf(xH) = \int_H f(xh)dxh$ is the averaging map $C_c(G) \rightarrow C_c(G/H)$, and Δ_G, Δ_H are the modular functions of G and of H .

Harmonic analysis

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theorem folland_2_51_invariant_measure_on_quotient {G : Type*} [Group G]
[TopologicalSpace G] [IsTopologicalGroup G]
[LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G]
(mu : Measure G) [mu.IsHaarMeasure]
(H : Subgroup G) (hH : IsClosed (H : Set G)) [LocallyCompactSpace H]
(nu : Measure H) [nu.IsHaarMeasure]
[TopologicalSpace (G / H)] [MeasurableSpace (G / H)] [BorelSpace (G / H)] :
(exists rho : Measure (G / H), rho != 0 /\
(forall g : G, rho.map (fun q => g * q) = rho) /\
forall f : G -> C, Continuous f -> HasCompactSupport f ->
integral x, f x dmu = integral q : G / H, (integral y : H, f
(Quotient.out q * y) dnu) drho) <=>
forall y : H, ((Measure.modularCharacterFun (y : G) : R>=0) : R) =
((Measure.modularCharacterFun y : R>=0) : R) := by
sorry
```

2.69 Theorem. On any locally compact group G we have $L^1(G)$ $L^p_p(G) = L^p(G)$ for $1 \leq p < \infty$. Moreover, $L^1(G) * L^\infty(G) = L^1(G) * C_{\{l\}}(G) = C_{\{l\}}(G)$ and $L^\infty(G) * L^1(G) = C_{\{r\}}(G) * L^1(G) = C_{\{r\}}(G)$. Notation: Throughout, G is a locally compact group with a fixed left Haar measure, $L_yf(x) = f(y^{-1}x)$ and $R_yf(x) = f(xy)$ are the left and right translates of f , and $f * g(x) = \int_G f(y)g(y^{-1}x)dy$ is convolution. $L^1(G)$ is a Banach algebra under convolution. $C_{\{l\}}(G)$ and $C_{\{r\}}(G)$ denote the bounded left- and right-uniformly continuous functions on G .

Harmonic analysis

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noncomputable def groupConv (mu : Measure G) (f g : G -> C) : G -> C :=
fun x => integral y, f y * g (y^-1 * x) dmu

theorem folland_2_69_convolution_factorization {G : Type*} [Group G]
[TopologicalSpace G] [IsTopologicalGroup G]
[LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G]
(mu : Measure G) [mu.IsHaarMeasure]
(p : R>=0inf) (hp : 1 <= p) (hp' : p != inf) (f : G -> C) (hf : MemLp f p mu) :
exists g h : G -> C, Integrable g mu /\ MemLp h p mu /\ forall^m x dmu,
groupConv mu g h x = f x := by
sorry
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4.43 Theorem. Suppose H is a closed subgroup of G . If f in $C_c(G)$, define F in $C_c(G/H)$ by $F(xH) = \int_H f(xy) dy$. Then $F = f|_{H^\perp}$, where we identify $(G/H)^\perp$ with $H^\perp$. If also $f|_{H^\perp}$ in $L^1(H^\perp)$, then (with Haar measures on H and $H^\perp$ suitably normalized) $\int_H f(xy) dy = \int_{H^\perp} f(x) dx$. Notation: G is a locally compact abelian group, G its dual group of continuous characters $\chi : G \rightarrow \mathbb{T}$, and $f(\chi) = \int_G f(x) \overline{\chi(x)} dx$ the Fourier transform. For a closed ideal I subset $L^1(G)$, $\text{nu}(I) = \{\chi : f(\chi) = 0 \text{ for all } f \in I\}$ is its cospectrum (hull) and, for E subset G , $\text{iota}(E) = \{f \in L^1(G) : f|_E = 0\}$ is the kernel of E ; $\text{nu}(f) := \text{nu}(\{f\})$. $H^\perp = \{\chi \in G : \langle y, \chi \rangle = 1 \text{ for all } y \in H\}$ is the annihilator of H .

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noncomputable def dualFourier (mu : Measure G) (f : G -> C) (xi : PontryaginDual G) : C :=
  integral x, f x * (starRingEnd C) ((xi x : C)) dmu

def annihilator (G : Type*) [CommGroup G] [TopologicalSpace G] (H : Subgroup G) :
  Subgroup G :=
  Subgroup (PontryaginDual G) where
    carrier := {xi | forall y in H, xi y = 1}
    one_mem' _ _ := rfl
    mul_mem' := fun {a b} hxi heta y hy => by
      rw [show (a * b) y = a y * b y from ContinuousMonoidHom.mul_apply a b y, hxi y hy, heta y hy, one_mul]
    inv_mem' := fun {a} hxi y hy => by
      rw [show a^-1 y = (a y)^-1 from rfl, hxi y hy, inv_one]

theorem folland_4_43_subgroup_fourier_formula {G : Type*} [CommGroup G]
  [TopologicalSpace G] [IsTopologicalGroup G]
  [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G]
  (mu : Measure G) [mu.IsHaarMeasure]
  (H : Subgroup G) (hH : IsClosed (H : Set G))
  [LocallyCompactSpace H] [MeasurableSpace (annihilator G H)]
  [BorelSpace (annihilator G H)] [LocallyCompactSpace (annihilator G H)] :
  exists (nu : Measure H) (sigma : Measure (annihilator G H)),
  nu.IsHaarMeasure /\ sigma.IsHaarMeasure /\
  forall f : G -> C, Continuous f -> HasCompactSupport f ->
  Integrable (fun xi : annihilator G H => dualFourier mu f xi) sigma
  ->
  forall x : G, integral y : H, f (x * y) dnu =
  integral xi : annihilator G H, dualFourier mu f xi * ((xi :
  PontryaginDual G) x : C) dsigma := by
  sorry
```

4.52 Theorem. If N subset G is closed, then $\text{nu}(\text{iota}(N)) = N$. Notation: G is a locally compact abelian group, G its dual group of continuous characters $\chi : G \rightarrow \mathbb{T}$, and $f(\chi) = \int_G f(x) \overline{\chi(x)} dx$ the Fourier transform. For a closed ideal I subset $L^1(G)$, $\text{nu}(I) = \{\chi : f(\chi) = 0 \text{ for all } f \in I\}$ is its cospectrum (hull) and, for E subset G , $\text{iota}(E) = \{f \in L^1(G) : f|_E = 0\}$ is the kernel of E ; $\text{nu}(f) := \text{nu}(\{f\})$.

Harmonic analysis

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noncomputable def hull (mu : Measure G) (I : Set (G -> C)) : Set
(PontryaginDual G) :=
  {xi | forall f in I, dualFourier mu f xi = 0}

noncomputable def kernel (mu : Measure G) (E : Set (PontryaginDual G)) :
Set (G -> C) :=
  {f | Integrable f mu /\ forall xi in E, dualFourier mu f xi = 0}

theorem folland_4_52_hull_of_kernel {G : Type*} [CommGroup G]
  [TopologicalSpace G] [IsTopologicalGroup G]
  [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G]
  (mu : Measure G) [mu.IsHaarMeasure]
  (N : Set (PontryaginDual G)) (hN : IsClosed N) : hull mu (kernel mu N)
  = N := by
  sorry
```

4.54 Theorem. If G is compact, then $\text{iota}(\text{nu}(I)) = I$ for every closed ideal I subset $L^1(G)$. Notation: G is a locally compact abelian group, G its dual group of continuous characters $\chi : G \rightarrow \mathbb{T}$, and $f(\chi) = \int_G f(x) \overline{\chi(x)} dx$ the Fourier transform. For a closed ideal I subset $L^1(G)$, $\text{nu}(I) = \{\chi : f(\chi) = 0 \text{ for all } f \in I\}$ is its cospectrum (hull) and, for E subset G , $\text{iota}(E) = \{f \in L^1(G) : f|_E = 0\}$ is the kernel of E ; $\text{nu}(f) := \text{nu}(\{f\})$.

Harmonic analysis

```
noncomputable def groupConv (mu : Measure G) (f g : G -> C) : G -> C :=
  fun x => integral y, f y * g (y^-1 * x) dmu

def IsLpClosed (p : R>=0inf) (mu : Measure G) (I : Set (G -> C)) : Prop :=
  forall f : G -> C, MemLp f p mu ->
  (forall eps : R>=0inf, 0 < eps -> exists g in I, MemLp g p mu /\
  elpNorm (f - g) p mu < eps) -> f in I

noncomputable def hull (mu : Measure G) (I : Set (G -> C)) : Set
(PontryaginDual G) :=
  {xi | forall f in I, dualFourier mu f xi = 0}

noncomputable def kernel (mu : Measure G) (E : Set (PontryaginDual G)) :
Set (G -> C) :=
  {f | Integrable f mu /\ forall xi in E, dualFourier mu f xi = 0}

theorem folland_4_54_spectral_synthesis_compact {G : Type*} [CommGroup G]
  [TopologicalSpace G] [IsTopologicalGroup G]
  [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G]
  (mu : Measure G) [mu.IsHaarMeasure] [CompactSpace G]
  (I : Submodule C (G -> C)) (hClosed : IsLpClosed 1 mu (I : Set (G ->
  C)))
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		<pre> (hmem : forall f in I, Integrable f mu) (hideal : forall g : G -> C, Integrable g mu -> forall f in I, groupConv mu g f in I) : kernel mu (hull mu I) = I := by sorry </pre>
4.55 Theorem. Let $G = \mathbb{R}^n$ with $n \geq 3$, and let S be the unit sphere in $\mathbb{R}^n$. There is a closed ideal I in $L^1(\mathbb{R}^n)$ such that $\text{nu}(I) = S$ but $I \neq \text{iota}(S)$. Notation: G is a locally compact abelian group, G its dual group of continuous characters $\text{xi} : G \rightarrow \mathbb{T}$, and $f(\text{xi}) = \int f(x) \text{conj}(\langle x, \text{xi} \rangle) dx$ the Fourier transform. For a closed ideal I subset $L^1(G)$, $\text{nu}(I) = \{\text{xi} : f(\text{xi}) = 0 \text{ for all } f \text{ in } I\}$ is its cospectrum (hull) and, for E subset G, $\text{iota}(E) = \{f \text{ in } L^1(G) : f _E = 0\}$ is the kernel of E; $\text{nu}(f) := \text{nu}(\{f\})$. On $\mathbb{R}^n$ the dual group is identified with $\mathbb{R}^n$ and $f(\text{xi}) = \int e^{i \langle x, \text{xi} \rangle} f(x) dx$.	Harmonic analysis	<pre> def IsLpClosed (p : ℝ>=0inf) (mu : Measure G) (I : Set (G -> ℂ)) : Prop := forall f : G -> ℂ, MemLp f p mu -> (forall eps : ℝ>=0inf, 0 < eps -> exists g in I, MemLp g p mu /\ eLpNorm (f - g) p mu < eps) -> f in I noncomputable def hull (mu : Measure G) (I : Set (G -> ℂ)) : Set (PontryaginDual G) := {xi forall f in I, dualFourier mu f xi = 0} noncomputable def kernel (mu : Measure G) (E : Set (PontryaginDual G)) : Set (G -> ℂ) := {f Integrable f mu /\ forall xi in E, dualFourier mu f xi = 0} theorem folland_4_55_schwartz_synthesis_failure {n : ℕ} (hn : 3 <= n) : exists I : Submodule ℂ (EuclideanSpace ℝ (Fin n) -> ℂ), IsLpClosed 1 volume (I : Set (EuclideanSpace ℝ (Fin n) -> ℂ)) /\ (forall f in I, Integrable f volume) /\ (forall (y : EuclideanSpace ℝ (Fin n)), forall f in I, (fun x => f (x - y)) in I) /\ {xi : EuclideanSpace ℝ (Fin n) forall f in I, F f xi = 0} = Metric.sphere (0 : EuclideanSpace ℝ (Fin n)) 1 /\ (I : Set (EuclideanSpace ℝ (Fin n) -> ℂ)) != {f : EuclideanSpace ℝ (Fin n) -> ℂ Integrable f volume /\ forall xi in Metric.sphere (0 : EuclideanSpace ℝ (Fin n)) 1, F f xi = 0} := by sorry </pre>
4.67 Theorem. Suppose I is a closed ideal in $L^1(G)$, f in $L^1(G)$, and $\text{nu}(f)$ superset $\text{nu}(I)$. If $\text{dnu}(I)$ inter $\text{dnu}(f)$ contains no nonempty perfect set, then f in I. Notation: G is a locally compact abelian group, G its dual group of continuous characters $\text{xi} : G \rightarrow \mathbb{T}$, and $f(\text{xi}) = \int f(x) \text{conj}(\langle x, \text{xi} \rangle) dx$ the Fourier transform. For a closed ideal I subset $L^1(G)$, $\text{nu}(I) = \{\text{xi} : f(\text{xi}) = 0 \text{ for all } f \text{ in } I\}$ is its cospectrum (hull) and, for E subset G, $\text{iota}(E) = \{f \text{ in } L^1(G) : f _E = 0\}$ is the kernel of E; $\text{nu}(f) := \text{nu}(\{f\})$. A set is "perfect" if it is closed and has no isolated points; $d E$ denotes the topological boundary of E in G.	Harmonic analysis	<pre> noncomputable def groupConv (mu : Measure G) (f g : G -> ℂ) : G -> ℂ := fun x => integral y, f y * g (y^-1 * x) dmu def IsLpClosed (p : ℝ>=0inf) (mu : Measure G) (I : Set (G -> ℂ)) : Prop := forall f : G -> ℂ, MemLp f p mu -> (forall eps : ℝ>=0inf, 0 < eps -> exists g in I, MemLp g p mu /\ eLpNorm (f - g) p mu < eps) -> f in I noncomputable def hull (mu : Measure G) (I : Set (G -> ℂ)) : Set (PontryaginDual G) := {xi forall f in I, dualFourier mu f xi = 0} theorem folland_4_67_synthesis_from_thin_boundary {G : Type*} [CommGroup G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] (I : Submodule ℂ (G -> ℂ)) (hclosed : IsLpClosed 1 mu (I : Set (G -> ℂ))) (hmem : forall f in I, Integrable f mu) (hideal : forall g : G -> ℂ, Integrable g mu -> forall f in I, groupConv mu g f in I) (f : G -> ℂ) (hf : Integrable f mu) (hspec : hull mu I subset hull mu {f}) (hthin : forall P : Set (PontryaginDual G), P.Nonempty -> Perfect P -> not P subset frontier (hull mu I) inter frontier (hull mu {f})) : f in I := by sorry </pre>
4.81 Theorem. If f is a bounded continuous function on G, the following are equivalent: a. f is the restriction to G of a continuous function on bG. b. f is the uniform limit of linear combinations of characters on G. c. f is uniformly almost periodic. Notation: G is a locally compact abelian group, G its dual group of continuous characters $\text{xi} : G \rightarrow \mathbb{T}$, and $f(\text{xi}) = \int f(x) \text{conj}(\langle x, \text{xi} \rangle) dx$ the Fourier transform. For a closed ideal I subset $L^1(G)$, $\text{nu}(I) = \{\text{xi} : f(\text{xi}) = 0 \text{ for all } f \text{ in } I\}$ is its cospectrum (hull) and, for E subset G, $\text{iota}(E) = \{f \text{ in } L^1(G) : f _E = 0\}$ is the kernel of E; $\text{nu}(f) :=$	Harmonic analysis	<pre> def IsUniformlyAlmostPeriodic (f : G -> ℂ) : Prop := forall eps : ℝ, 0 < eps -> exists s : Finset G, forall y : G, exists z in s, forall x : G, rightTranslate y f x - rightTranslate z f x < eps theorem folland_4_81_almost_periodic_characterization {G : Type*} [CommGroup G] [TopologicalSpace G] [IsTopologicalGroup G] [LocallyCompactSpace G] [MeasurableSpace G] [BorelSpace G] (mu : Measure G) [mu.IsHaarMeasure] </pre>

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nu({f}). bG is the Bohr compactification of G, and f is *uniformly almost periodic* when the set of its right translates {R_yf : y in G} is totally bounded in the uniform norm.		(f : G -> C) (hf : Continuous f) (hbdd : exists C : R, forall x, f x <= C) : (forall eps : R, 0 < eps -> exists (s : Finset (PontryaginDual G)) (c : PontryaginDual G -> C), forall x, f x - sum xi in s, c xi * (xi x : C) < eps) <=> IsUniformlyAlmostPeriodic f := by sorry